Homework 3

Definitions

Define the following terms/concepts

- a.) Sampling Error
- b.) Law of Large Numbers
- c.) Sample Space
- d.) Event
- e.) Union
- f.) Intersection
- g.) Random Trial
- h.) Random Variable
- i.) Probability distribution
- j.) Population distribution
- k.) Continuous random variable
- l.) Discrete random variable
- m.) Probability Mass Function
- n.) Probability Density Function

Simple probability

(**1**) A fair coin is thrown 3 times. What is the probability that at least one heads is observed?

(**2**) Suppose that in a class of 30 students, 8 students are in band, 15 students play a sport, and 5 students are both in band and play a sport. Let A be the event that a student is in band and let B be the event that a student plays a sport. Create a Venn diagram that models this situation. What is the probability that a student is in the band or plays a sport?

(**3**) A standard deck of 52 cards contains four suits (Hearts, Diamonds, Clubs, and Spades). Each suit contains 13 cards (9 numbered cards and 3 face cards, and an Ace).

- Suppose a single card is drawn at random, what is the probability that it is a numbered card?
- What is the probability that the selected card is the Ace of Spades (assume independence where necessary)?

(**4**) An urn contains 5 green balls, 7 red balls, and 4 yellow balls. Two balls are selected at random with replacement. What is the probability that first ball is yellow and the second ball is red?

(**5**) The probability that it rains R on a given day in Moscow, Idaho is $P(R) = 0.1$. The probability that it is Windy W is $P(W) = 0.2$, and the probability that it is cloudy C is $P(C) = 0.7$. Assume that these three types of weather events are independent. What is the probability that a given day is Rainy, Cloudy, and Windy?

Sampling Distributions, Means, and Variances

(**6**) A random variable X has the following probability distribution

X	P(X)
1	0.25
2	0.25
3	0.25
4	0.25

- Derive the sampling distribution for the mean of $n = 2$ observations of X
- Compute the mean $\mu_{\bar{X}}$ and standard deviation $\sigma_{\bar{X}}$ of the sampling distribution of $\bar{X}$

(**7**) A random variable Y has two possible outcomes. The outcome $Y = 1$ occurs with probability $p = 0.23$. Derive the sampling distribution for the proportion of outcomes with $Y = 1$ in a sample of $n = 3$ observations of Y.

(**8**) Consider a survey of tree volume. Suppose that in the population of trees 10% have a volume of 10 cubic feet, 20% have a volume of 20 cubic feet, 30% have a volume of 30 cubic feet, and 40% have a volume of 40 cubic feet. Let X denote the volume of a randomly selected tree, assuming that every tree has an equal chance of being selected. The probability distribution of X is given in the table below.

X	P(X)
10	0.1
20	0.2
30	0.3
40	0.4

- Confirm that X has a mean of $\mu = 30$ cubic feet and a standard deviation of $\sigma = 10$ cubic feet by computing them yourself.
- Derive the sampling distribution for the mean tree volume in a sample of $n = 2$ trees.

Deriving Probabilities From Discrete Distributions

(**9**) Michael Dickson, who plays for the Seattle Seahawks, is a punter in the NFL. The following sampling distribution gives the mean punting distance (in yards) for a sample of $n = 4$ punts. Use this distribution to answer the following questions:

$\bar{X} =$ Mean Punt Distance	$P(\bar{X})$
35.33	0.40
45.67	0.50
55.00	0.08
60.33	0.01
65.67	0.01

- What is the probability the mean distance is 55 or more yards?
- What is the probability the mean distance is at least 35.33 yards?
- What is the probability the mean distance is between 35.33 and 45.67 yards?
- What is the probability the mean distance is 35.33 yards or 55 yards?

(**10**) A random variable X follows a binomial distribution with probability of success $p = 0.43$ and number of trials $n = 8$. What is the probability of observing fewer than 3 successes, that is what is $P(X < 3)$?

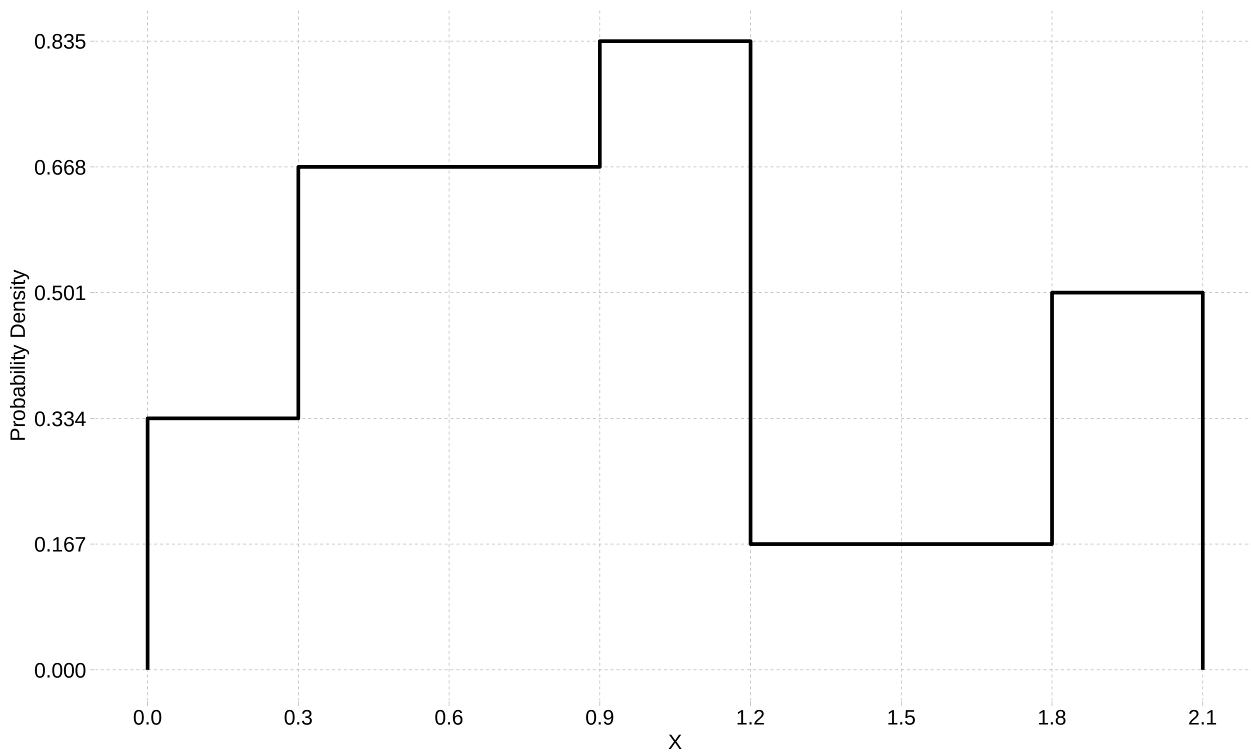
(**11**) Tom Brady is a famous quarterback who recently retired from playing in the National Football League. During his career, the probability that Brady would complete a pass in a given play was 64.3%. If a typical NFL team runs about 35 offensive pass plays, what is the probability that Brady would complete exactly 28 pass plays in a game?

(**12**) A random variable W follows a Poisson distribution with rate parameter $\lambda = 1.8$ events per hour. What is the probability of observing exactly four events in a given hour, that is what is $P(W = 4)$?

(**13**) A factory experiences on average 3 equipment malfunctions per day, following a Poisson distribution with a rate parameter of 3. Calculate the probability that on a randomly selected day, the factory will observe 3 or more equipment malfunctions.

Deriving Probabilities From Continuous Distributions

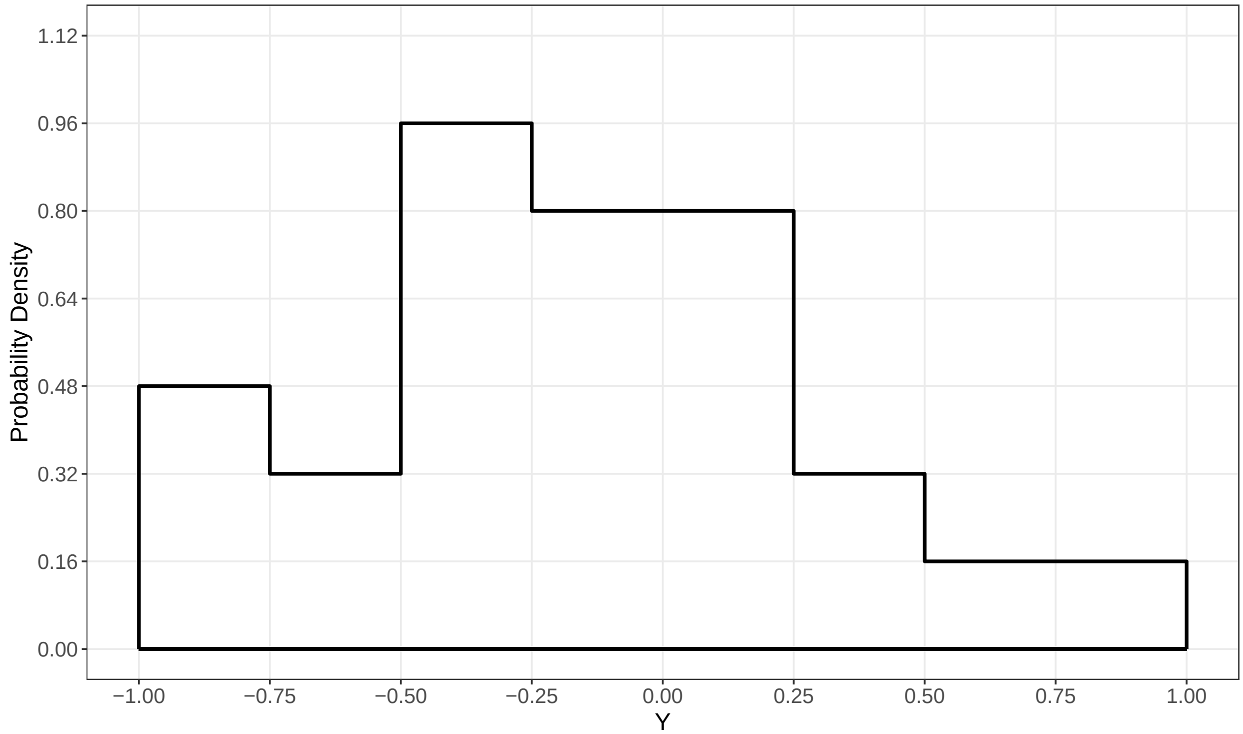
(**14**) Use the plot below of the probability distribution of a continuous random variable X to answer the following questions



Use the plot to find the following probabilities:

- What is $P(X < 0.6)$?
- What is $P(X > 0.3)$?
- What is $P(1.5 < X < 1.8)$?

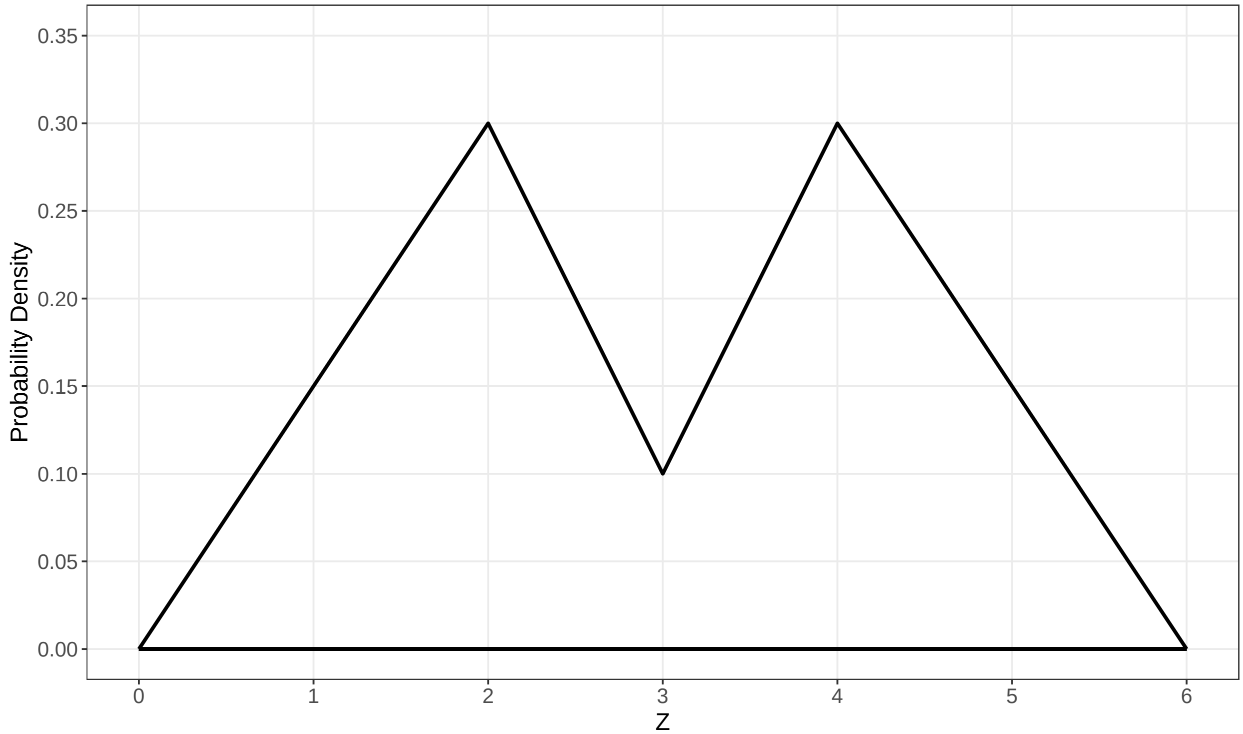
(**15**) Use the plot below of the probability distribution of a continuous random variable Y to answer the following questions



Use the plot to find the following probabilities (recall that the area of a triangle is $\frac{1}{2} \times$ base $\times$ height)

- What is $P(Y > -0.25)$?
- What is $P(-0.5 < Y < 0.25)$?

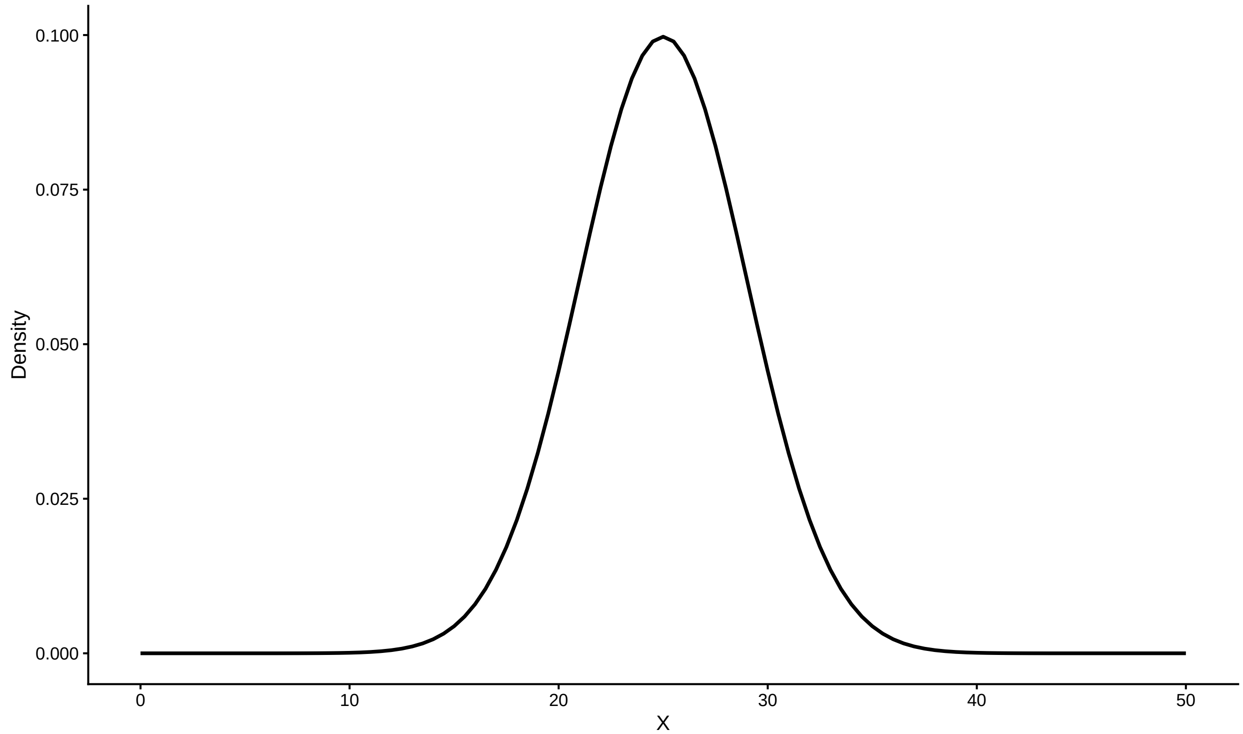
(**16**) Use the plot below of the probability distribution of a continuous random variable Z to answer the following questions



Use the plot to find the following probabilities (recall that the area of a triangle is $\frac{1}{2} \times$ base $\times$ height)

- What is $P(Z < 2)$?
- What is $P(2 < Z < 3)$?
- What is $P(Z > 4)$?
- Give the approximate value of the $25th$ percentile of distribution of Z

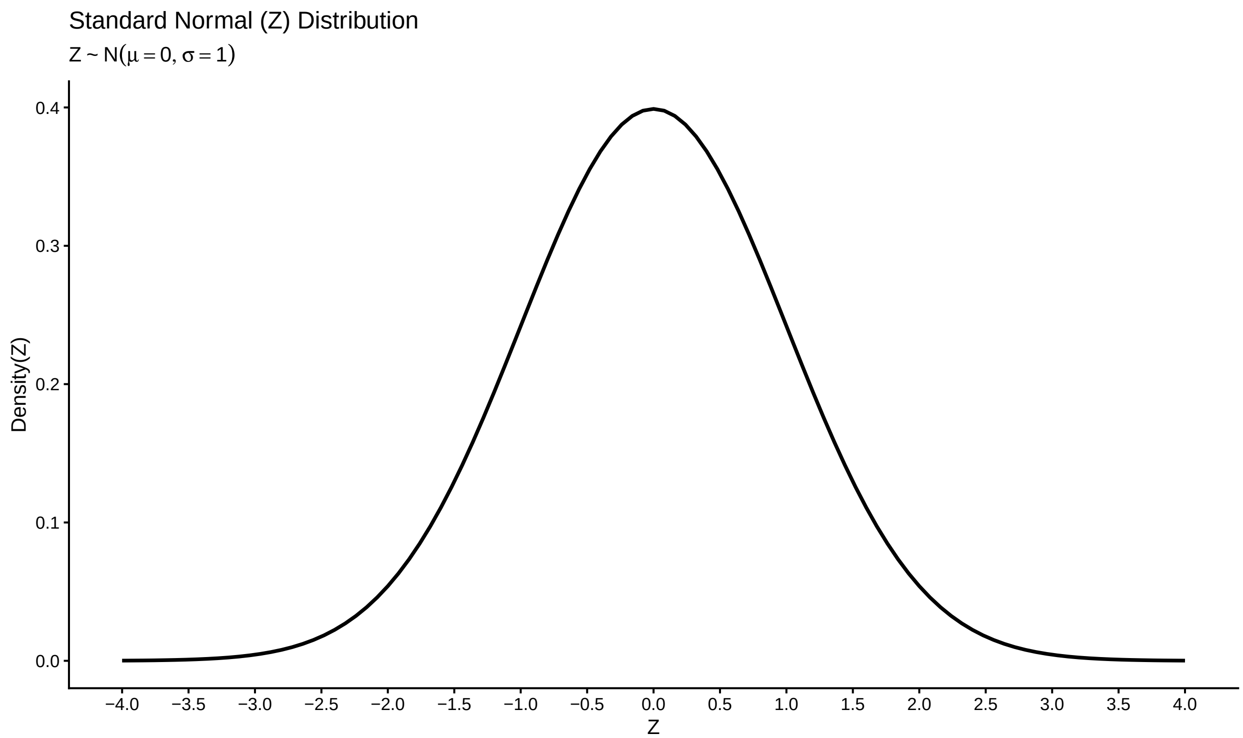
17.) Assume the continuous random variable X follows a normal distribution with mean $\mu = 25$ and variance $\sigma^2 = 16$.



Using a Z-table, find the following (approximate) probabilities for X. You can also use the app in the course website or StatDistributions.com to check your result

- What is $P(X < 15)$?
- What is $P(X > 28)$?
- What is $P(12 < X < 20)$?
- What is $P(X > 30)$?
- What is $P(25 < X < 33)$?
- What is the value of X that marks the $95th$ percentile of the distribution?
- What is the value of X that marks the $40th$ percentile of the distribution?

18.) Suppose that a random variable X follows a normal distribution with mean $\mu = -5$ and standard deviation $\sigma = 1.8$. Using the plot of the Z-distribution below, shade in the area corresponding to the probability $P(-2 < X < 0.5)$


Standard Normal (Z) Distribution
Z ~ N(μ = 0, σ = 1)
Density(Z)
0.4
0.3
0.2
0.1
0.0
−4.0
−3.5
−3.0
−2.5
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Z

References